

Nile University (NU) IoT Research Internship Report

Day 9 & 10

Overview

Days 9 and 10 are treated as one continuous unit, focused on establishing a Wi-SUN (Wireless Smart Ubiquitous Network) connection using three CC1352P7 LaunchPad kits. The `ns_node` example from the SimpleLink SDK was used on all three kits to form a Wi-SUN mesh network, a key technology behind smart city and industrial IoT applications.

Task 1: Project Setup and Firmware Flashing

Configured all three CC1352P7 LaunchPad kits with the `ns_node` example from the SimpleLink SDK, imported via Code Composer Studio from the Wi-SUN examples path. Built the project and flashed the `ns_node` firmware onto each of the three kits in turn.

This was the setup step for the Wi-SUN work, following the same pattern used for BLE (Day 5) and Zigbee (Days 6-7): before any network behavior could be tested, each board needed the correct example firmware loaded through Code Composer Studio.

Task 2: Wi-SUN FAN Mesh Network Formation

Established a Wi-SUN FAN (Field Area Network) mesh connection between the three devices, with each kit running as a router node. Verified through Putty debug output that each device scanned for the network, authenticated successfully, and joined with a unique device address, forming a connected mesh where Kit 2 discovered the network via Kit 1, and Kit 3 discovered it via Kit 2.

Improvement over Task 1: Task 1 only got the correct firmware running on each board individually. This task proved the boards could actually find each other and form a working mesh network, extending each device's role from a standalone router node into an active member of a connected topology.

Task 3: Communication Testing Between Devices

Tested communication between the devices in the established mesh network and confirmed that all three kits were actively routing packets, as shown in the debug output ("Routing packets active" on each device).

Task 4: Understanding the Router Node Role

Studied the router node role within the Wi-SUN mesh topology, including the NWP (Network Processor) architecture used by `ns_node`, where router nodes provide upward and downward packet forwarding and their interconnections form the overall mesh.

What is Wi-SUN?

Wi-SUN is a protocol based on IEEE 802.15.4 that enables large-scale mesh networks with long-range communication, low power consumption, and self-healing capabilities, where devices automatically reroute around failed nodes. It is widely used in smart city applications such as street lighting, utility metering, and environmental monitoring.

ns_node Example Overview

The ns_node example implements a router node using NWP (Network Processor) mode architecture. Router nodes can connect to other devices in the mesh and provide both upward and downward packet forwarding; the connections between router nodes form the overall mesh topology.

Key characteristics:

- Device support: CC1312R1, CC1352P-1, CC1352R1, and CC1352P7 LaunchPads
- Memory footprint: approximately 15kB flash and 14kB RAM
- Role: router node with NWP architecture
- Protocol: Wi-SUN FAN (Field Area Network)

Hardware Setup



Three CC1352P7 LaunchPad kits were used, each connected via USB for power and programming, with a computer running Code Composer Studio for flashing. All three kits (Kit 1, Kit 2, Kit 3) were configured identically as router nodes, each connected to the network and capable of routing packets.

Setup and Configuration

The wi-sun/ns_node project was imported into Code Composer Studio from the SimpleLink SDK examples path for the CC1352P7 LaunchPad, built, and flashed to each of the three kits in turn. As with previous days, Putty was used over UART (115200 baud, 8 data bits, 1 stop bit, no parity, no flow control) to view Wi-SUN network debug messages for each device, confirming network discovery, authentication, and successful joining.

Key Learnings

Wi-SUN concepts:

- FAN (Field Area Network): a large-scale mesh network
- Router Node: a device that can forward packets to other devices
- Border Router: a gateway between the Wi-SUN network and external networks
- Mesh topology: a self-healing network where devices can reroute around failed nodes
- NWP mode: Network Processor mode, where the host manages communication

CC1352P7 capabilities:

- Sub-1GHz communication for long-range wireless connectivity
- Multi-protocol support: Wi-SUN, Zigbee, Bluetooth Low Energy, and Thread
- Integrated power amplifier with up to 20dBm output power
- Low power design, suitable for battery-operated devices

Tools & Technologies

- IDE: Code Composer Studio (TI)
- SDK: SimpleLink CC13xx/CC26xx SDK
- Protocol: Wi-SUN FAN (IEEE 802.15.4)
- Hardware: TI CC1352P7 LaunchPad (3 kits)
- Debug: Putty (serial terminal)
- MCU: ARM Cortex-M4F

Real-World Applications

- Smart cities: street lighting, traffic management
- Utility metering: smart meters for water, gas, and electricity
- Industrial IoT: machine monitoring and process control

- Environmental monitoring: air quality and weather stations

Outcome

- Installed and configured the SimpleLink SDK with Wi-SUN support
- Flashed the ns_node firmware on three CC1352P7 LaunchPad kits
- Established a working Wi-SUN mesh network connection between all three devices
- Verified network formation, device joining, and active packet routing
- Understood the router node role and NWP architecture within Wi-SUN mesh topology
- Completed the internship's progression across wireless protocols: BLE (Day 5, two devices)
→ Zigbee (Days 6-7, three-device network) → Wi-SUN (Days 9-10, three-device mesh with self-healing routing)